

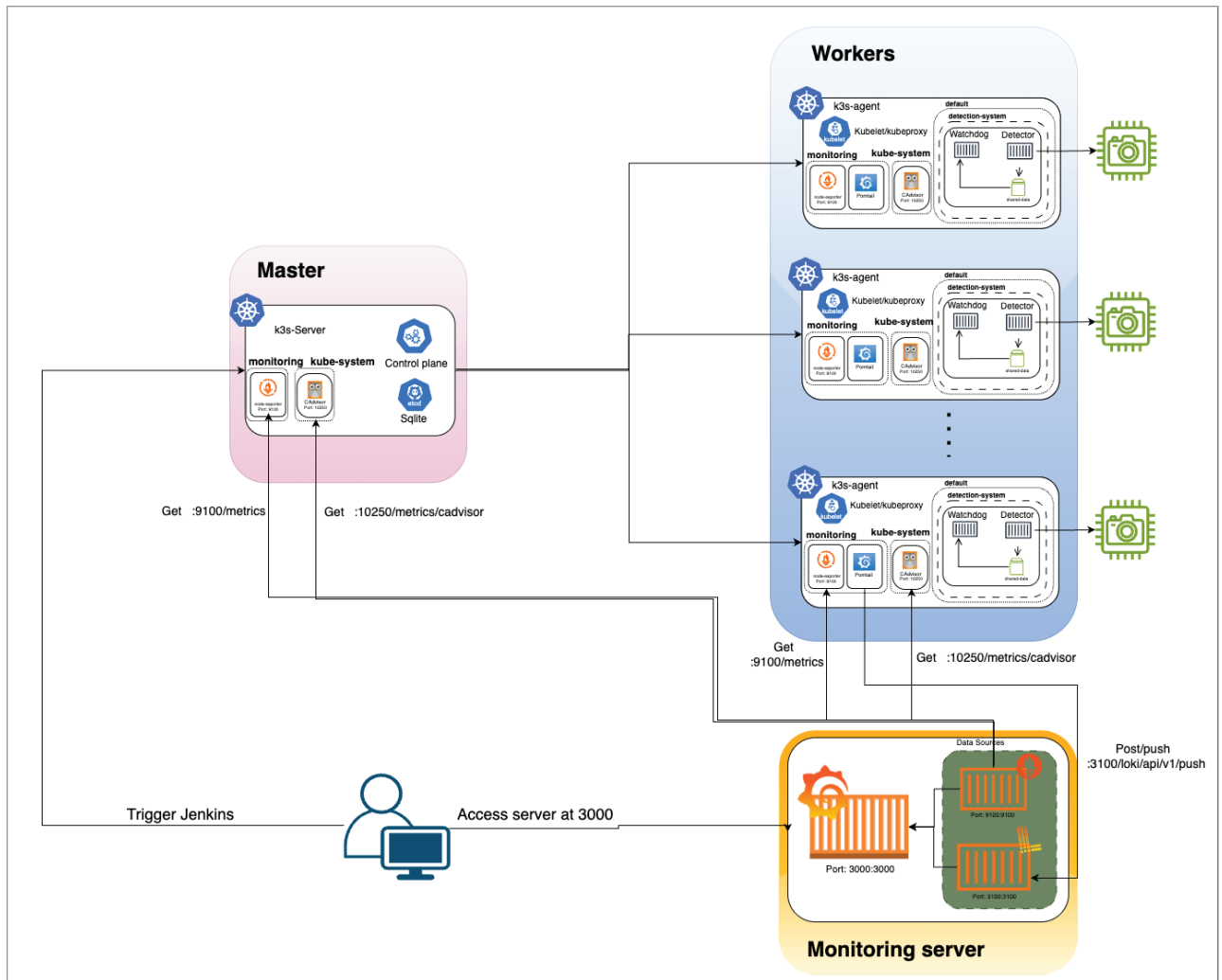


Figure 1: Integrated edge orchestration architecture showing component relationships

and after the event) to a shared volume and to S3 for archival storage.

The Watchdog container continuously monitors the shared volume for new clips. In our current prototype, it polls the directory every 3 seconds (using Python's watchdog or a similar mechanism), balancing responsiveness with low CPU overhead. Upon finding a new clip, Watchdog sends an email alert (via SMTP) with the video attached and then deletes the clip to conserve space. This loop provides near-realtime alerts with low latency. Infrastructure automation is achieved via Ansible playbooks: each Pi node's `/boot/config.txt` is tuned for GPU

memory allocation (`gpu_mem=128`) and its CPU frequency governor set to performance for maximum throughput. A Jenkins pipeline automates Docker image builds and tagging; it pushes and triggers the Ansible playbooks to update the cluster. Security is enforced with Kubernetes role-based access control (RBAC) and Jenkins credentials management system, with sensitive credentials for AWS/S3 and SMTP passed securely to Ansible using the `--extra-vars` parameter during deployment.

Centralised monitoring is provided by an on-premises observability stack. The local server runs Prometheus, Loki,

and Grafana. Each edge node hosts a Prometheus Node Exporter (metrics on port 9100) and cAdvisor (container stats on port 10250). A Promtail agent on each Pi collects system logs and forwards them to the central Loki instance (port 3100). Prometheus scrapes the Node Exporter and cAdvisor endpoints at 15-second intervals, while Grafana (port 3000) provides real-time dashboards of node health (CPU/GPU/memory utilisation), model inference latency, and alert events. This unified stack provides visibility into system performance without relying on external cloud services, aligning with the hybrid cloud-edge design.